Boolean Algebra And Logic Circuit

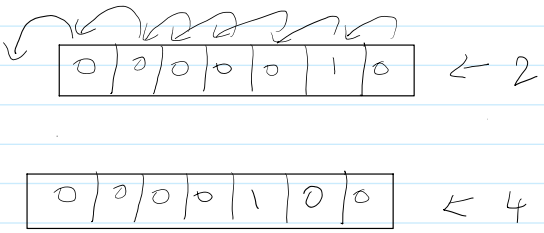
12 September 2026 09:58

The binary operations performed by any digital circuit with the set of elements 0 and 1 are called logical operations or logical functions. The algebra used to symbolically represents the logic function called Boolean algebra. It is a two state algebra invented by a mathematician George Boole in 1854.

Thus, the Boolean algebra is a system of mathematical logic for analysis and designing of digital systems.

A variable or function of variables in Boolean algebra can assume only 2 values, either 0 or 1. Hence there are no fraction, no negative number, no square root, no cube root, no logarithms, etc.

- **Logical Operation**
  - **Computer can perform only three kinds of operation.**
    - **Addition of bits.**
    - **Shifting of bits (left shift and right shift.)**
    - **Complements of bits.**



1. Logical addition:-OR

0 + 0 = 0
0 + 1 = 1
1 + 0 = 1
1 + 1 = 1

← can be compared with normal arithmetic

Boolean logic of addition of 2-bit is, if any of the bit is 1, then result is 1

2. Logical Multiplication:- AND

0.0 = 0
0.1 = 0
1.0 = 0
1.1 = 1

Boolean logic of multiplication states that, when both bits are '1', then result is '1', otherwise result is '0'

3. Complement:- NOT

0' = 1
1' = 0

4. Laws of Boolean Algebra

a. Properties of '0'
i. 0 + p = p
ii. 0 . p = 0

b. Properties of '1'
i. 1 + p = 1
ii. 1.p = p

c. Absorption properties
i. p + pq = p
ii. p + (p + q) = p + q

d. Involution property
i. (p')' = p

e. Idempotence property
- **i. p + p = p**
- **ii. p.p = p**

f. Complementarity Property
- **i. p + p' = 1**
- **ii. p.p' = 0**

g. Commutative property
- **i. p + q = q + p**
- **ii. p.q = q.p**

h. Associative property
- **i. (p + q) + r = p + (q + r)**
- **ii. (p.q).r = p.(q.r)**

i. Distributive property
- **i. p(q + r) = p.q + p.r**
- **ii. p + q.r = (p + q)(p + r)**
- **iii. p + p'q = (p + q)**

j. De'Morgan's law or Properties
- **i. (p + q)' = p'.q'**
- **ii. (p.q)' = p' + q'**

What is Truth Table?
A truth table is a mathematical table used in logic to represent all the combination of truth values for all variables involved in boolean expression.

	p	q	$(p+q)'$	p'	q'	$p'.q'$	$(p.q)'$	$p'+q'$
1	0	0	1	1	1	1	1	1
2	0	1	0	1	0	0	1	1
3	1	0	0	0	1	0	1	1
4	1	1	0	0	0	0	0	0

1st law = 2nd law Hence Proved

Proof of DeMorgan's theorem using Truth Table.

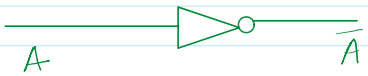
Logic gates
A gate is simply an electronic circuit which operates on one or more signals to produce an output signal. Gates are digital (2 - state) circuit because the input and output signals are either zero (low voltage) or one (high voltage) Gates are often called logic circuit because they can be analyzed with Boolean algebra.

Logic gates are of 3 types:
1. Basic gates
- **i. NOT gate**
- **ii. OR Gate**
- **iii. AND Gate**

2. Universal gates
   i. NOR Gate or NOT(OR) Gate
   ii. NAND Gate or NOT(AND) Gate
3. Exclusive gates
   i. XOR Gate or EXclusive OR Gate
   ii. XNOR Gate or NOT(XOR) Gate

Basic Gates

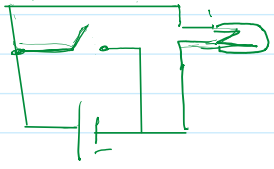
1. NOT Gate: Is the gate with only one input signal and 1 inverted output signal or state is always opposite or barred.



Logical symbol

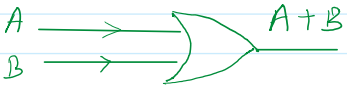
A	$\bar{A}$
0	1
1	0

Truth table

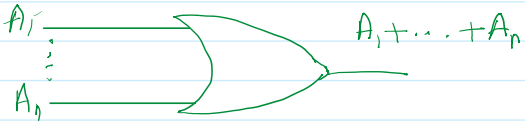


ON ← Not glow
OFF ← glow
Electrical ckt.

2. OR Gate: The OR gate has two or more input signals, but only one output signal. If the input signal is one or high or TRUE, the output signal is 1 or high or true. Otherwise it is 0 or low or false. It is represented with plus(+) symbol.

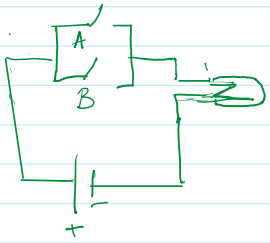


Two-input OR gate



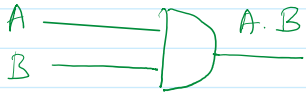
More than 2-input OR Gate

	A	B	A+B
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

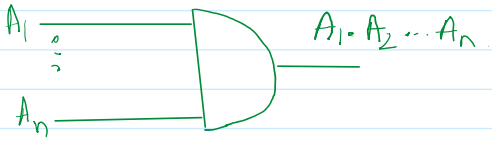


Truth table for OR gate 2-inputs & one output

3. AND Gate: The AND gate can have two or more input signal and produce an output signal. When all the input signals are one or high or true then output signal is 1 or high or true, otherwise zero or low or false. It is Represented by dot '.' symbol.

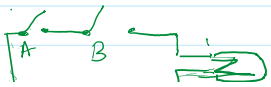


Two-input AND gate

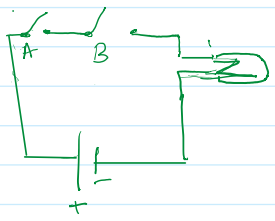


N-input AND gate

	A	B	A.B
0	0	0	0
1	0	1	0
2	1	0	0



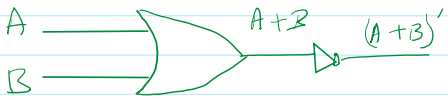
1	0	0	0
2	0	1	0
3	1	0	0
4	1	1	1



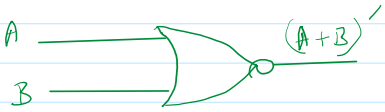
Truth table for AND gate 2-inputs & one output

Universal Gates:

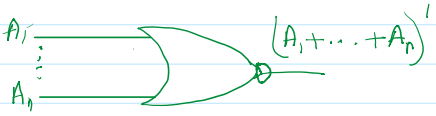
1. NOR Gate: In Nor gate we can input two or more signals, but we will get only one output signal. The output of NOR gate is inverted OR gate.



NOT(OR) gate (two input)



NOR gate (Two input)



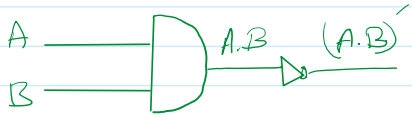
NOR gate (N-input)

$$(A+B)' = A'B'$$ ← De Morgan's law.

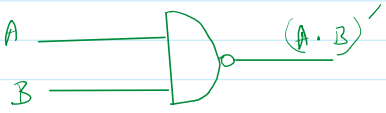
	A	B	A'	B'	(A+B)'	A'B'
1	0	0	1	1	1	1
2	0	1	1	0	0	0
3	1	0	0	1	0	0
4	1	1	0	0	0	0

Truth table for 2-input NOR gate

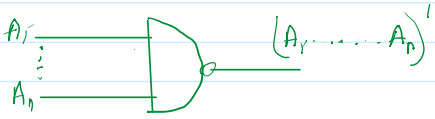
2. NAND Gate: In NAND gate we can input two or more signal but we can get only one output signal. In NAND gate we get the output of inverted AND gate.



NOT(AND) gate (two input)



NAND gate (Two input)



NAND gate (N-input)

$$(A \cdot B)' = A' + B'$$ ← De Morgan's law.

	A	B	A'	B'	(A.B)'	A'+B'
1	0	0	1	1	1	1
2	0	1	1	0	1	1

	A	B	A′	B′	(A.B)′	A′+B′
0	0	0	1	1	1	1
1	0	1	1	0	1	1
2	1	0	0	1	1	1
3	1	1	0	0	0	0

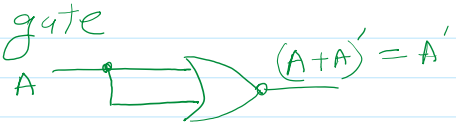
Truth table for 2-input NAND gate.

Why we use Universal Gates?
Ans: A universal gate is a logic gate which can implement any boolean function without using any other type of logic gate. The NOR gate and NAND gates are universal gates. This means you can create any logical boolean expression using only NOR gates or only NAND gates. We call it:

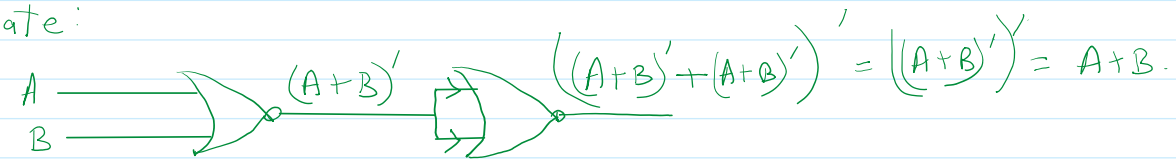
- NOR - to - NOR logic
- NAND - to - NAND logic

- **NOR - to - NOR Logic**

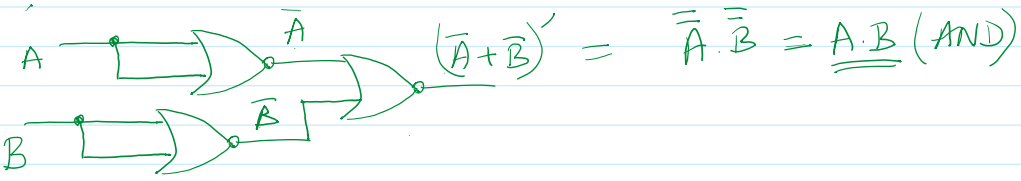
NOT gate



OR gate:

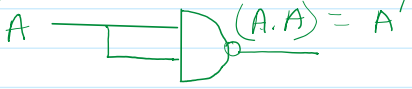


AND gate

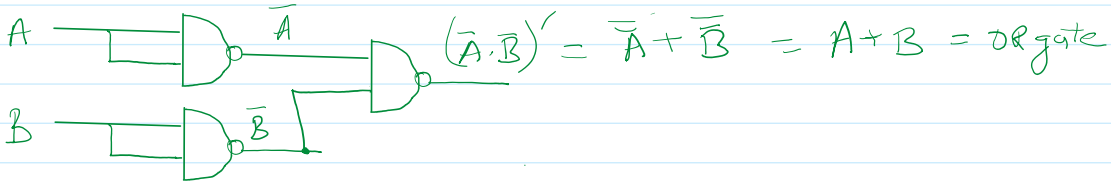


- **NAND - to - NAND logic**

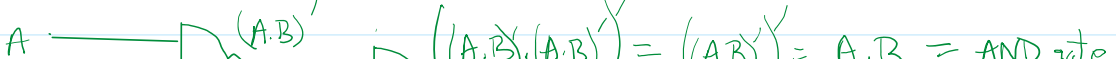
NOT Gate:



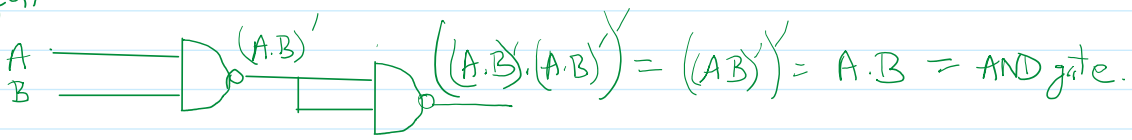
OR Gate:



AND gate:



AND gate:

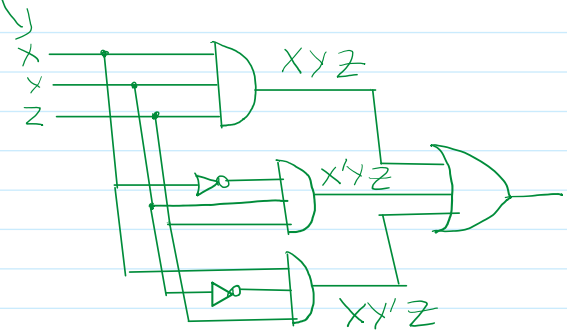


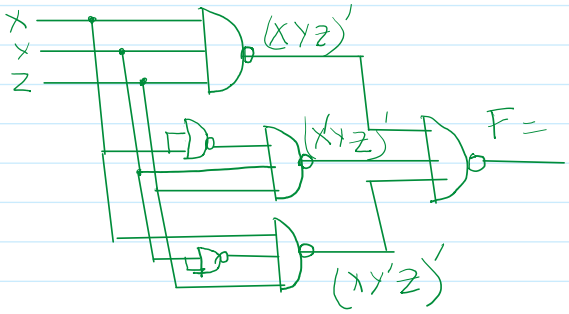
Design the circuit using NAND gates only or NOR gates only:

- **Draw the expression using basic gates only.**
- **Replace all the gates with NAND gates or NOR gates depending upon a question.**
- **Try to resolve the expression, if you are getting the correct result, no need to add NAND gate or NOR gate to adjust the answer.**
- **If the expression is given in the form of SOP(Sum of Products), use NAND gates to draw the circuit, and**
- **If the expression is given in the form of POS(Product of Sum), use NOR gate to draw the circuit.**

SOP expression F(X, Y, Z) = XYZ + X'YZ + XY'Z [Use NAND Gate]

POS expression F(X, Y, Z) = (X+Y+Z).(X'+Y+Z).(X+Y'+Z) [use NOR Gate]





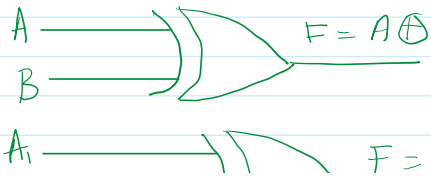
$= [(XYZ)'.(X'YZ)'.(XY'Z)']'$

$= ((XYZ)')' + ((X'YZ)')' + ((XY'Z)')'$

$= XYZ + X'YZ + XY'Z$

Exclusive Gates:

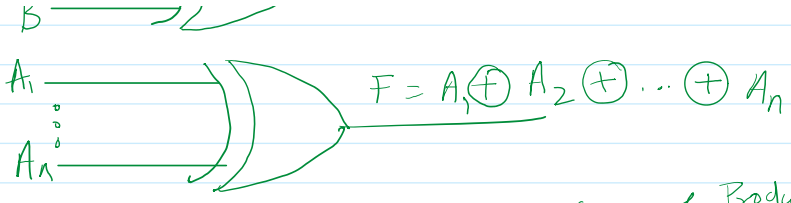
1. XOR : It takes 2 or more input signals and output one signal. The output signal is high or 1 or true, iff we had odd number of high or 1 or true input signals in a row, otherwise output is low, 0 or false.
It is represented by '⊕' plus in a circle symbol.



A	B	F
1	0	0
0	1	1

✓ (A+B)

✓ A'B



A	B	F	
0	0	0	✓ (A+B)
0	1	1	✓ A'B
1	0	1	✓ AB'
1	1	0	✓ (A'+B')

Truth Table

Sum of Product

SOP

'1', '0' barred.

$F = A \oplus B \oplus C$

A	B	C	F	
0	0	0	0	✓
0	0	1	1	✓
0	1	0	1	✓
0	1	1	0	✓
1	0	0	1	✓
1	0	1	0	✓
1	1	0	0	✓
1	1	1	1	✓

minterms

$A \oplus B = A'B + AB'$

Product of sum

POS

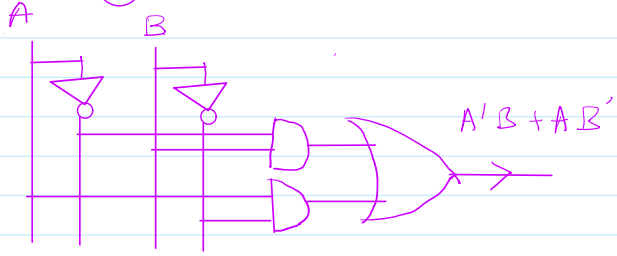
'0', '1' is barred

$A \oplus B = (A+B)(A'+B')$

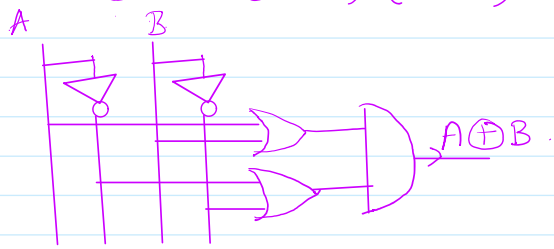
$SOP \rightarrow (A \oplus B \oplus C) = A'B'C + A'BC' + AB'C' + ABC$

$POS \rightarrow (A \oplus B \oplus C) = (A+B+C)(A+B'+C')(A'+B+C')(A'+B'+C)$

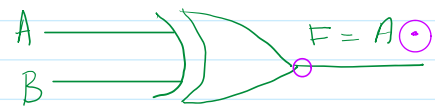
SOP $A \oplus B = A'B + AB'$

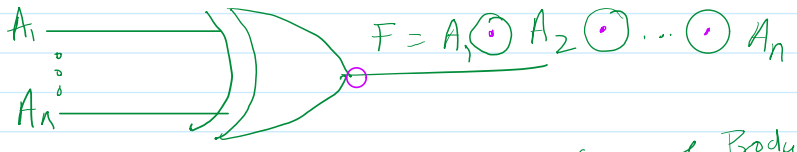


POS $A \oplus B = (A+B)(A'+B')$



XNOR : It takes 2 or more input signals and output one signal. The output signal is high or 1 or true, iff we had even number of high or 1 or true input signals in a row, otherwise output is low, 0 or false.
It is represented by '⊙' dot in a circle symbol.





Sum of Product

A	B	F	
0	0	1	A'B'
0	1	0	(A+B')
1	0	0	(A'+B)
1	1	1	AB

Truth Table

A' ——

1	1	1

AB

Truth Table

Sum of Product

SOP

'1', '0' barred.

minterms

$A \odot B = A'B + AB$

$F = A \odot B \odot C$

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Product of sum

POS

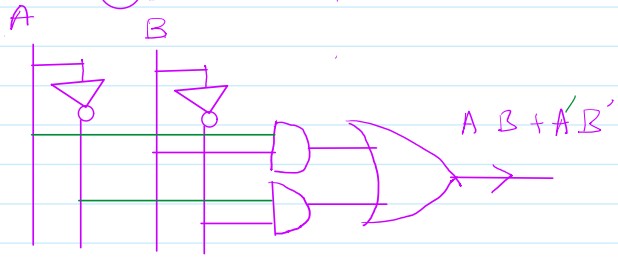
'0', '1' is barred

$A \odot B = (A+B)(A'+B')$

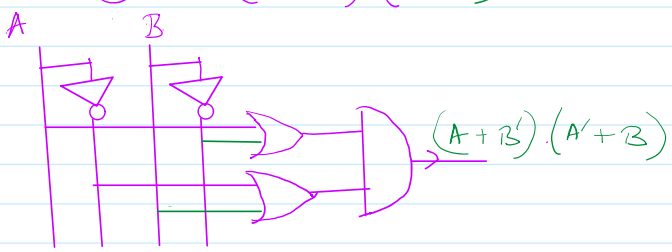
$SOP \rightarrow (A \odot B \odot C) = A'B'C + A'BC' + AB'C' + ABC$

$POS \rightarrow (A \odot B \odot C) = (A+B+C)(A+B'+C')(A'+B+C')(A'+B'+C)$

SOP $A \odot B = AB + A'B'$



POS $A \odot B = (A+B')(A'+B)$



H.W Draw the ckt by using basic gates of 3-input of XNOR gate.

BASIC DUALITY IN BOOLEAN ALGEBRA

We state the duality theorem in the following:

1. Changing each OR (+) sign to an AND (·) sign.
2. Changing each AND (·) sign to an OR (+) sign.
3. Complenting each ('0' to 1) or ('1' to 0)

$A + 0 = A$

$A \cdot 1 = A$

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

dual relationship $A + B \cdot C = (A + B) \cdot (A + C)$

} distributive law

dual ↔ $(A + B)' = A' \cdot B'$

dual ↔ $(A \cdot B)' = A' + B'$

dual